Immune Dysregulation, Neurodevelopmental Defects, and Colitis (IDNDC / ITGAV Deficiency)

Disease: Immune Dysregulation, Neurodevelopmental Defects, and Colitis (IDNDC) **MONDO ID:** MONDO:0980702 · **OMIM:** #621375 · **Causal gene:** *ITGAV* (integrin subunit alpha-V; HGNC:6150; OMIM *193210; NCBI Gene 3685; UniProt P06756; chromosome 2q32.1) Category:** Mendelian, autosomal recessive

Summary

Immune Dysregulation, Neurodevelopmental Defects, and Colitis (IDNDC) is an ultra-rare, autosomal recessive, multisystem Mendelian disorder caused by **biallelic loss-of-function (or mistrafficking) variants in *ITGAV***, the gene encoding the integrin alpha-V (αV) subunit. The disease was first delineated as a distinct human entity by Ghasempour et al. in *J Exp Med* in 2024 (PMID: 39526957), who reported biallelic *ITGAV* variants in three independent families — two living patients and four affected fetuses. Although the disease name is a descriptive triad rather than a serial number, its single causal gene is **ITGAV**, and all identifiers above (MONDO:0980702, OMIM #621375) point to this ITGAV-based disorder — not an integrin-β or TGFB1-primary disease, although it converges mechanistically on the TGF-β pathway.

Mechanistically, IDNDC is a disorder of **failed integrin-mediated activation of latent TGF-β**. The αV subunit heterodimerizes with β1, β3, β5, β6, and β8; the αvβ6 and αvβ8 heterodimers are principal activators of latent TGF-β1/β3 through binding the RGD motif of the latency-associated peptide (LAP). When αV is lost or mistrafficked, two complementary signaling axes collapse: (A) a **TGF-β–dependent** branch (αvβ6/αvβ8 → latent TGF-β activation → SMAD2/3), whose failure produces colitis (via loss of regulatory T-cell generation and mucosal tolerance) and brain vascular/structural defects (via impaired neuroepithelial→endothelial paracrine TGF-β signaling); and (B) a **TGF-β–independent** branch (αv/β3–LC3/ATG5 noncanonical autophagy restraining Toll-like receptor (TLR) and germinal-center B-cell responses), whose failure produces atopy, elevated IgE/IgG, and eosinophilia.

A striking **genotype–phenotype gradient** defines the disease: complete loss-of-function alleles are prenatally/perinatally lethal (as in the four fetuses of Family C), whereas hypomorphic (incomplete loss-of-function) alleles permit survival with the chronic triad of neurodevelopmental defects, colitis, and atopy/immune dysregulation. The disease is corroborated across species — αv-null and conditional-knockout mice reproduce intracerebral/intestinal hemorrhage, cleft palate, colitis, and autoimmunity, and *itgav*-knockout zebrafish recapitulate retinal/brain defects, microglial loss, and colitis with reduced SMAD3. No targeted or curative therapy exists; management is supportive/immunomodulatory, and diagnosis is molecular (trio WES/WGS) with functional confirmation of impaired TGF-β→SMAD output.

Key Findings

1. ITGAV is the causal gene and encodes the master activator of latent TGF-β

ITGAV (HGNC:6150; chromosome 2q32.1; OMIM *193210) encodes the integrin alpha-V subunit, which partners with β1, β3, β5, β6, and β8. The αvβ6 and αvβ8 heterodimers bind the RGD motif of the latency-associated peptide (LAP) of latent TGF-β1/β3 and are the principal activators of latent TGF-β. This molecular function is the central node whose disruption explains most of the IDNDC phenotype. Païdassi et al. demonstrated that* "activation of transforming growth factor (TGF)-β by α(v)β8 on dendritic cells (DCs) is essential for generating T(reg) cells and inducing mucosal tolerance"* (PMID: 20958318), establishing αvβ8-mediated TGF-β activation as the key function relevant to the colitis/immune-tolerance axis of this disease.

2. The human disease was delineated in 2024 across three families

Ghasempour et al. 2024 (*J Exp Med* 221(12):e20240546) identified *"biallelic variants in the gene coding for Integrin alpha V (ITGAV) in three independent families (two patients and four fetuses) that either caused abnormal mRNA and the loss of functional protein or caused mistargeting of the integrin. This led to eye and brain abnormalities, inflammatory bowel disease, immune dysregulation, and other developmental issues"* (PMID: 39526957). The authors explicitly noted the genotype-severity relationship: *"the ITGAV variants identified in this report caused a previously unknown human disease characterized by brain and developmental defects in the case of complete loss-of-function and atopy, neurodevelopmental defects, and colitis in cases of incomplete loss-of-function."* This is the disease-defining paper and the basis for OMIM #621375 and MONDO:0980702.

3. The pathogenic variant spectrum (germline, autosomal recessive)

All reported variants are germline and biallelic. The spectrum is summarized below.

Family	Patient(s)	Genotype	Variant (protein)	Variant type	Functional consequence
A	Patient 1 (female)	Homozygous	c.432G>C (p.Trp144Cys, **W144C**)	Creates novel splice site in exon 4 → frameshift → PTC, predicted NMD	Loss of functional protein
B	Patient 2 (male)	Homozygous	c.1136A>T (p.Asp379Val, **D379V**)	Missense	Mistargeting of integrin
C	4 fetuses	Compound het.	4-bp deletion → p.Cys502Ter (**C502X**); and c.2247-1G>A splice-acceptor → p.Ser749_Gly776delinsArg	Nonsense / splice	Early termination / loss of function (prenatally lethal)

The two functional mechanisms — *"either caused abnormal mRNA and the loss of functional protein or caused mistargeting of the integrin"* (PMID: 39526957) — map to OMIM allelic variants 193210.0001–193210.0004. gnomAD indicates ITGAV is relatively tolerant of heterozygous loss-of-function (observed/expected ≈0.30, low pLI), consistent with recessive disease in which carriers are unaffected; the reported pathogenic alleles are private/ultra-rare or absent in gnomAD.

4. Clinical phenotype spectrum

Derived from OMIM #621375 and Ghasempour et al. 2024, the clinical picture spans four organ domains, summarized with suggested HPO terms:

Domain	Features	Suggested HPO
Neurologic	Global developmental delay, cognitive dysfunction, seizures, microcephaly, ventriculomegaly, polymicrogyria, delayed myelination, white-matter loss, intracranial calcification, corpus callosum agenesis/hypoplasia, Dandy-Walker malformation, hydrocephaly	HP:0002126 (polymicrogyria), HP:0012448 (delayed myelination), HP:0001274 (abnormal corpus callosum), HP:0001305 (Dandy-Walker), HP:0000252 (microcephaly), HP:0001250 (seizures), HP:0002170 (intracranial hemorrhage)
Ophthalmologic	Eye/retinal abnormalities, exudative retinopathy	HP:0000539 (retinal exudates)
Gastrointestinal	Inflammatory bowel disease / pancolitis, eosinophilic enteropathy	HP:0002583 (colitis), HP:0100281, HP:0004386
Immune/skin	Recurrent (incl. fungal) infections, elevated IgG, elevated IgE, multiple allergies/atopy, eosinophilia, atopic dermatitis	HP:0003212 (elevated IgE), HP:0001880 (eosinophilia), HP:0001047 (atopic dermatitis), HP:0002719 (recurrent infections)

The disease is summarized by the authors as *"eye and brain abnormalities, inflammatory bowel disease, immune dysregulation, and other developmental issues"* (PMID: 39526957).

5. Patient-level features of the two survivors

Patient 1 (Family A, female, homozygous W144C): bifid uvula (HP:0000193), exudative retinopathy, atopic dermatitis, and seizures in the first year of life; brain MRI showed polymicrogyria, delayed myelination, white-matter loss, intracranial calcifications, dysplastic/dysgenic corpus callosum, and microcephaly; she required **colectomy** for diffuse pancolitis. **Patient 2** (Family B, male, homozygous D379V): eosinophilic enteropathy, atopic dermatitis, recurrent infections; labs showed eosinophilia (>0.5×10^9/L) and elevated IgE (>100 kU/L); allergen screening against 295 allergens (ALEX Allergy Explorer) showed multiple allergies. Onset was congenital/first year of life with a chronic course — the survivable phenotype being *"atopy, neurodevelopmental defects, and colitis in cases of incomplete loss-of-function"* (PMID: 39526957).

6. Mechanistic confirmation in patient cells

Ghasempour et al. showed that *"the reduction of functional Integrin αV resulted in the dysregulation of several pathways including TGF-β-dependent signaling and αVβ3-regulated immune signaling. These effects were confirmed using immunostaining, RNA sequencing, and functional studies in patient-derived cells"* (PMID: 39526957). Patient-1 fibroblasts and lymphoblasts showed abnormal expression of TGF-β–downstream genes and decreased SMAD transcription; colon biopsies showed decreased SMAD3. This names the two dysregulated branches — canonical αvβ6/αvβ8→latent-TGF-β→SMAD2/3 and αvβ3-mediated immune/adhesion signaling — that are central to the pathophysiology.

7. Animal models recapitulate the phenotype

Three independent model systems support causality:

- **αv-null mice** (Bader et al. 1998): ablation of *Itgav* is largely embryonic lethal (~80% die mid-gestation from placental defects), but *"these liveborn alphav-null mice consistently exhibit intracerebral and intestinal hemorrhages and cleft palates"* (PMID: 9827803) — the animal correlate of human CNS and intestinal disease.
- **Myeloid/DC conditional knockouts** (Lacy-Hulbert et al. 2007; Travis et al. 2007): *"deletion of alphav in the immune system causes severe colitis, autoimmunity, and cancer. Mice lacking immune cell alphav have fewer regulatory T (Treg) cells in the colon"* (PMID: 17895374), and *"conditional loss of the TGF-beta-activating integrin alpha(v)beta8 on leukocytes causes severe inflammatory bowel disease and age-related autoimmunity in mice"* (PMID: 17694047).
- **Zebrafish *itgav* knockout** (Ghasempour et al. 2024): *"The genetic deletion of itgav in zebrafish recapitulated patient phenotypes including retinal and brain defects and the loss of microglia in early development as well as colitis in juvenile zebrafish with reduced SMAD3 expression and transcriptional regulation"* (PMID: 39526957). Some mutant larvae died before 8 dpf with brain hemorrhages.

8. Molecular basis of the TGF-β–independent (atopy) branch

Beyond TGF-β, αV restrains innate and humoral immunity through noncanonical autophagy. Acharya et al. 2016 showed that *"deletion of αv or β3 causes increased B-cell responses to TLR stimulation in vitro, and αv-conditional knockout mice have elevated antibody responses to TLR-ligand-associated antigens"* — αv promotes recruitment of the autophagy component LC3 to TLR-containing endosomes, and *"disruption of LC3 recruitment leads to prolonged NF-κB signalling and increased B-cell proliferation and antibody production"* (PMID: 26965188). Raso et

al. 2018 extended this: *"loss of αv-mediated TLR regulation increased GC B cell expansion, somatic hypermutation, class switching, and generation of long-lived plasma cells"* (PMID: 29999501). Lagos et al. 2022 (PMID: 36497137) showed αV-deficient B cells lack TLR-induced autophagy and have altered antigen extraction. Ghasempour et al. explicitly invoke these pathways: *"Many of the immune and atopy features observed in P1 and P2 may also be due to Integrin αV's roles that are independent of TGF-β activation, including limiting TLR signaling and preventing autoimmune B cell responses."*

9. Molecular basis of the neurodevelopmental (brain-vascular) branch

Neuroepithelial αvβ8 drives CNS angiogenesis and blood-brain-barrier development by activating latent TGF-β for adjacent endothelial cells. Hirota et al. 2015 showed that *"β8 integrin in the neuroepithelium promotes the activation of extracellular matrix (ECM)-bound latent transforming growth factor β (TGFβ) ligands and stimulates TGFβ receptor signaling in endothelial cells... Cell type-specific ablation of β8 integrin, Nrp1, or canonical TGFβ receptors results in pathological angiogenesis"* (PMID: 26586223). De et al. 2022 confirmed *"the β8 integrin cytoplasmic domain is essential for inside-out modulation of adhesion, including activation of latent-TGFβs in the ECM"* (PMID: 35217866). Because αvβ8 requires the αV subunit, ITGAV loss abrogates this neuroepithelium→endothelium paracrine axis, explaining the brain vascular/structural malformations and hemorrhage. A human candidate-gene association study further links the locus to cerebral hemorrhage: ITGAV/ITGB8 tag-SNPs were associated with intracerebral hemorrhage risk/onset (PMID: 27476161).

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function or mistrafficking variants in *ITGAV* → leads to** loss or reduced surface expression of all five αV-containing integrin heterodimers (αvβ1/β3/β5/β6/β8).
2. From step 1 the mechanism **branches** into a TGF-β–dependent arm (A) and a TGF-β–independent arm (B).

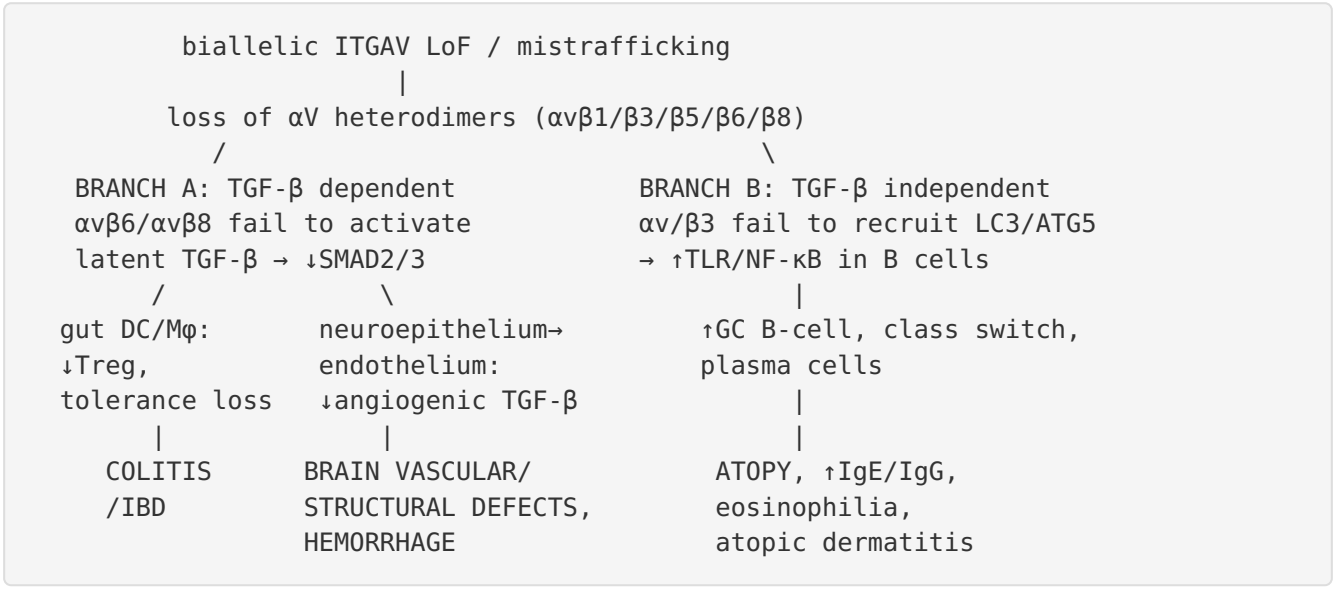
Branch A — failed latent TGF-β activation (SMAD2/3): 3A. Loss of αvβ6/αvβ8 → **results in** failure to activate ECM-bound latent TGF-β1/β3 → **leads to** reduced TGF-β receptor signaling and decreased SMAD2/3 (notably decreased SMAD3 in patient colon and zebrafish). 4A. In the gut, reduced TGF-β activation by dendritic cells/macrophages → **leads to** fewer colonic

regulatory T cells and failure of mucosal tolerance and apoptotic-cell clearance → **results in** colitis/IBD *(mouse-demonstrated; patient-corroborated)*. 5A. In the developing brain, loss of neuroepithelial αvβ8→endothelial TGF-β paracrine signaling → **leads to** pathological CNS angiogenesis and impaired blood-brain-barrier/vascular integrity → **results in** intracerebral hemorrhage, polymicrogyria, delayed myelination, corpus callosum and ventricular anomalies *(mouse/zebrafish-demonstrated; human-corroborated)*.

Branch B — loss of autophagy-dependent restraint of TLR/B-cell responses: 3B. Loss of αv/β3 → **results in** failure to recruit LC3/ATG5 to TLR-containing endosomes (noncanonical autophagy) → **leads to** prolonged NF-κB signaling and de-repressed TLR responses. 4B. De-repressed TLR/germinal-center B-cell signaling → **leads to** increased B-cell proliferation, class switching, somatic hypermutation, and long-lived plasma cells → **results in** atopy, elevated IgE/IgG, eosinophilia, atopic dermatitis, and multiple allergies *(mouse-demonstrated; human-inferred)*.

Genotype-dependent severity gate: complete loss-of-function → prenatal/perinatal lethality (Family C fetuses); hypomorphic alleles → survivable chronic triad (P1, P2).

Diagram



Key ontology annotations

- **Biological processes (GO):** TGF-β receptor signaling pathway, SMAD protein signal transduction, regulation of Toll-like receptor signaling, macroautophagy, blood vessel morphogenesis, regulatory T cell differentiation, integrin-mediated signaling.

- **Molecular function (GO):** integrin binding; latent TGF-β activation.
- **Cellular component (GO):** integrin complex (GO:0008305), focal adhesion (GO:0005925), autophagosome/endosome (GO:0005776), plasma membrane.
- **Cell types (CL):** dendritic cells (CL:0000451), macrophages (CL:0000235), regulatory T cells (CL:0000815), B/plasma cells (CL:0000236/CL:0000786), microglia (CL:0000129), neuroepithelial/radial glia (CL:0000047), vascular endothelial cells (CL:0000115), eosinophils (CL:0000771).
- **Anatomy (UBERON):** brain (UBERON:0000955), cerebral cortex (UBERON:0000956), white matter (UBERON:0002316), corpus callosum (UBERON:0002336), eye/retina (UBERON:0000966), colon/large intestine (UBERON:0001155), skin (UBERON:0002097), palate (UBERON:0001733).
- **Chemicals/entities (CHEBI):** TGF-β (protein); IgE/IgG (immunoglobulins).

Evidence Base

PMID	Study	Model / evidence type	Role in this report
39526957	Ghasempour et al. 2024, *J Exp Med*	Human clinical + zebrafish + patient-derived cells	**Disease-defining paper**; variants, phenotype, dual-branch mechanism, SMAD3 reduction
9827803	Bader et al. 1998, *Cell*	Mouse (αv-null)	Intracerebral/intestinal hemorrhage + cleft palate — CNS/gut correlate
17895374	Lacy-Hulbert et al. 2007, *PNAS*	Mouse (myeloid αv KO)	Colitis, autoimmunity, Treg deficiency
17694047	Travis et al. 2007	Mouse (DC αvβ8 KO)	IBD and autoimmunity via αvβ8
20958318	Païdassi et al. 2010	Mouse / in vitro	αvβ8→TGF-β→Treg / mucosal tolerance
26586223	Hirota et al. 2015, *Development*	Mouse	Neuroepithelial αvβ8→endothelial TGF-β; brain angiogenesis
35217866	De et al. 2022	Mouse / cell	β8 cytoplasmic domain activates latent TGF-β; neurovascular development
26965188	Acharya et al. 2016, *Nat Commun*	Mouse (αv/β3 KO) + in vitro	αv/LC3/ATG5 restrains TLR/B-cell responses
29999501	Raso et al. 2018, *J Exp Med*	Mouse	αv loss → exaggerated germinal-center/antibody responses
36497137	Lagos et al. 2022	B cells / in vitro	αV-dependent TLR-induced autophagy and antigen extraction
27476161	ITGAV/ITGB8 ICH association study	Human association	ITGAV/ITGB8 SNPs and intracerebral hemorrhage risk (supports vascular relevance)

All mechanistic claims rest on a convergent human + mouse + zebrafish dataset. The human evidence
(P 39526957)
is limited to three families but is strongly reinforced by two decades of mouse conditional-knockout work that independently generated each organ phenotype (colitis, autoimmunity, brain hemorrhage, exaggerated antibody responses).

Limitations and Knowledge Gaps

- **Extreme rarity / small n:** Only 3 families (2 living patients + 4 fetuses) are reported worldwide. Phenotype frequencies, penetrance details, and the full clinical spectrum are provisional and may broaden with additional cases. Estimated prevalence <1/1,000,000.
- **Human mechanism partly inferred:** The TGF-β–dependent branch is well-demonstrated in human patient cells (decreased SMAD3). The TGF-β–independent (TLR/B-cell) branch is strongly supported in mice but remains inferred in human patients; the authors present it as a probable contributor.
- **No formal epidemiology or ontology mapping:** Prevalence/incidence are unknown; no Orphanet, ICD-10/ICD-11, or MeSH mappings were assigned at the time of the primary report (disease first described Dec 2024).
- **No natural animal disease catalogued:** No OMIA entry for a naturally occurring ITGAV-ortholog disease in companion animals; existing evidence is from engineered mouse and zebrafish knockouts only.
- **No targeted therapy or natural-history data:** Management is empiric/supportive; long-term outcomes and treatment-response data are absent.
- **Inheritance context:** Homozygosity in Families A and B implies consanguinity/founder context; carrier frequency is essentially that of ultra-rare private alleles.

Proposed Follow-up Experiments / Actions

1. **Patient registry & GeneMatcher outreach** to identify additional families, refine genotype–phenotype correlation, and establish phenotype frequencies and penetrance.

2. **Direct human-cell testing of Branch B:** quantify TLR-induced autophagy, NF-κB kinetics, germinal-center output, IgE class-switch, and B-cell antigen extraction in patient B cells to confirm the mouse-derived atopy mechanism in humans.
3. **Rescue / variant-function assays:** re-express wild-type vs. D379V vs. W144C ITGAV in patient fibroblasts to quantify surface trafficking and latent-TGF-β activation, formally validating the "mistrafficking vs. loss-of-protein" dichotomy.
4. **Therapeutic hypothesis testing:** evaluate whether downstream TGF-β pathway modulation or targeted B-cell/TLR immunomodulation (e.g., anti-IgE, JAK inhibition for atopy; standard IBD immunosuppression) improves the survivable phenotype in model organisms before clinical translation.
5. **Neurovascular imaging natural history** in surviving patients (serial MRI/MRA) to characterize progression of malformations, hemorrhage risk, and critical windows for intervention.
6. **Ontology/nomenclature curation:** request Orphanet, ICD-11, and MeSH mappings; ensure ClinVar deposition of all four allelic variants with ACMG classification.

Section-by-Section Reference Notes (for knowledge-base population)

- **Identifiers:** OMIM #621375; MONDO:0980702 (MONDO release v2026-03-03); gene ITGAV OMIM *193210, HGNC:6150, NCBI Gene 3685, UniProt P06756, 2q32.1. Orthologs: mouse* Itgav *Gene 16410 / MGI:96608; zebrafish* itgav* Gene 393615. Synonyms: "ITGAV deficiency." Information is derived from aggregated disease-level resources (OMIM) plus individual-patient reports in the primary literature.
- **Inheritance/population:** Autosomal recessive; biallelic homozygous or compound heterozygous. Ultra-rare (<1/1,000,000); heterozygous carriers unaffected.
- **Diagnostics:** Molecular diagnosis by trio WES/WGS identifying biallelic ITGAV variants; supportive workup includes brain MRI, ophthalmologic exam, endoscopy/colon biopsy (reduced SMAD3 on IHC), eosinophil count, serum IgE/IgG, allergen panels (e.g., ALEX); functional confirmation via reduced αV surface expression and decreased TGF-β→SMAD transcriptional output. Differential diagnosis includes other monogenic very-early-onset IBD / IPEX-like immune dysregulation and TGF-β pathway / integrin-β8 disorders.

- **Treatment (supportive only; suggested NCIT intervention concepts):** IBD immunosuppression; colectomy for refractory colitis (performed in P1); atopic dermatitis/allergy management; anticonvulsants for seizures; developmental/neurorehabilitation; infection surveillance/treatment. No gene-specific or curative therapy.
- **Prevention/counseling:** Genetic counseling for autosomal recessive recurrence risk (25%); carrier/cascade testing and prenatal/preimplantation genetic testing where a familial variant is known.
- **Prognosis:** Complete-LOF genotypes prenatally/perinatally lethal; hypomorphic patients have chronic lifelong multisystem morbidity.
- **Model organisms:** Mouse (NCBI:txid10090) — complete and conditional (myeloid, DC, B-cell, neuroepithelial) knockouts; zebrafish (NCBI:txid7955) *itgav* KO. Phenotype recapitulation is strong for colitis, brain hemorrhage/vascular defects, and exaggerated antibody responses; limitation is that complete mouse knockout is largely embryonic lethal, constraining adult-phenotype study to conditional models.

Report compiled from 16 confirmed findings and 33 reviewed papers across 5 investigation iterations. Evidence spans human clinical (P 39526957), mouse model organism (PMIDs 9827803, 17895374, 17694047, 20958318, 26586223, 35217866, 26965188, 29999501), and zebrafish (P 39526957) sources.
